MOSTAFA SAMEER

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I am committed to advancing resilient, data-driven infrastructure decision-making and am currently seeking opportunities to apply my analytical and transportation engineering skills to real-world infrastructure challenges.

ACADEMIC QUALIFICATION

- **Ph.D. in Civil Engineering** (Aug 2024 – **Aug 2028**): Florida International University (FIU)
  Advisor: **Dr. Wallied Orabi,** Associate Professor, Moss School of Construction, FIU. GPA: **3.90** out of 4.00
  Focus: Developing an interpretable framework for bridge rehabilitation prioritization through the combination of structural conditions, transportation network efficiency, and community accessibility.
- **M.Sc. in Civil Engineering – Transportation Engineering** (Sep 2023 – Aug 2024): Wayne State University (WSU)
  Advisor: Dr. Xiaodong Qian, Assistant Professor, CEE, WSU. GPA: 3.47 out of 4.00
- **M.Sc. in Civil Engineering – Transportation Engineering** (August 2023 – May 2024): King Fahd University of Petroleum and Minerals (KFUPM) – Transferred to Wayne State University and continued my master's there.
  Advisor: Dr. Hassan M. Al-Ahmadi, Associate Professor of Civil Engineering, King Fahd University of Petroleum & Minerals. CGPA: 3.00 out of 4.00
- **MBA – Operations & Supply Chain Management** (Jan 2017 – Sep 2019): University of Chittagong, Bangladesh
  Advisor: Dr. Sagib Kumar Ghosh, Professor, Marketing, University of Chittagong. CGPA: 2.99 out of 4.00
- **B.Sc. in Civil Engineering** (Oct 2011 – Jan 2016): Ahsanullah University of Science & Technology, Dhaka
  Advisor: Dr. Md. Shahid Mamun, Professor, AUST. Thesis: Determination of Trip Attraction Rates of Different Size Hospitals in Dhaka City.

EXPERIENCE (RESEARCH/ WORK)

Graduate Research Assistant (Jan 2026 – Present), **FIU – Moss School of Construction**
- Developing GIS-based coastal exposure/risk workflows to quantify resilience across multiple spatial resolutions.
- Developing an interpretable framework for bridge rehabilitation prioritization through the combination of structural conditions, transportation network efficiency, and community accessibility**.**

Graduate Research Assistant – Digital Twin Lab (Aug 2024 – Dec 2025)
- Investigated hurricane-induced wind-driven rain intrusion through sliding glass door systems in partnership with the Florida Building Commission
- Co-authored technical report: Standards review for wind-driven rain intrusion through sliding glass door tracks (DOI: 10.13140/RG.2.2.21522.16322).

Quality Assurance Engineer / Technician (Co-Op) (Jun 2024 – Aug 2024), **Fluor Corporation – Gordie Howe International Bridge, Detroit, MI**
- Executed QA/QC procedures for structural elements per ASTM and AASHTO standards on a USD 5.6B binational megaproject,
- Managed compliance documentation, slump testing, air density pressure testing, and field coordination with testing stakeholders.

Graduate Teaching Assistant – Emerging Mobility Lab (Sep 2023 – May 2024), **Wayne State University**
- Developed Python-driven simulation experiments in CARLA/Unreal Engine 4 to evaluate traffic optimization strategies,
- Modeled V2X assumptions for e-scooters and mixed traffic networks; assessed efficiency and safety trade-offs.

Graduate Teaching Assistant (Aug 2022 – May 2023), **King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia**
- Taught 85+ undergraduate students in Drafting and Transportation Engineering; designed network models in Synchro, VISSIM, AutoTURN.

Civil Engineer (Nov 2019 – Jul 2022), **KDS Textile Mills Ltd., Chattogram, Bangladesh**
- Managed factory load distribution and related structural considerations to support building safety and structural integrity
- Applied engineering software including AutoCAD, ETABS, STAAD.Pro, GIS, and Civil 3D for analysis, design, and project support.
- Supervised construction activities associated with industrial facilities and coordinated engineering requirements for ongoing projects.

Senior Officer, Civil Construction (Mar 2017 – Oct 2019),**Youngone Sports Shoes Industries Ltd., Chattogram, Bangladesh**
- Monitored and maintained the structural integrity of factory buildings and supported compliance with applicable safety requirements.
- Prepared structural integrity reports documenting building conditions and identified issues requiring corrective action.

Assistant Engineer (Feb 2016 – Jan 2017), **Luminars Engineers Association, Chattogram, Bangladesh**
- Conducted structural and building inspections to assess existing conditions and support retrofit and rehabilitation activities.

- Assisted in evaluating the structural integrity of existing buildings and documenting engineering findings.

PUBLICATIONS

Journal Publications

- Gazder, U., Islam, M. K., **Sameer, M.*** (2026). Long-Term Transportation Fuel Consumption in Annex I Countries: A Mixed-Method Multi-Sector Analysis (1990–2020). https://doi.org/10.32604/ee.2026.086249
- Aldossari, A., Alhajji, B., Hanbzaza, R., Haque, M. E., Islam, M. K., & **Sameer, M.*** (2026). Data-Driven Decarbonization: Clustering of National CO_2 Emission Profiles and Economic Indicators for Sustainable Energy Transitions. *Energy Engineering*, SI: Science, Engineering, and Policy Innovations Driving the Global Energy Transition. Issue 88402. https://doi.org/10.32604/ee.2026.088402
- Islam, M. K., Hassan, M., **Sameer, M.**, & Gazder, U. (2026). Energy and Emission Implications of Electric and Autonomous Vehicle Deployment in Saudi Arabia's Urban Transport Systems. https://doi.org/10.1109/FET68771.2026.11601878
- Islam, M. K., & **Sameer, M.** (2026). Advancing Sustainable Mobility in GCC Countries: A Pathway to Achieving Sustainable Development Goals. Proceedings of the 3rd International Conference on Sustainability. https://doi.org/10.1007/978-981-92-1726-7_17
- Hassan, M., Kabir, M. E., Islam, M. K., Rambe, A. H., Jusoh, M., & **Sameer, M.** (2025). Mapping the Machine Learning Landscape in Autonomous Vehicles: A Scientometric Review of Research Trends, Applications, Challenges, and Future Directions. IEEE Access, 13, 182036–182077. https://doi.org/10.1109/ACCESS.2025.3620637
- Zaman, T. A. B., Haque, T., Sen Gupta, H., **Sameer, M.**, & Nofal, O. (2025). Computational Review of Disaster Resilience Research in Bangladesh. Progress in Disaster Science (Elsevier). https://doi.org/10.1016/j.pdisas.2025.100486
- Mahin, H. D., Hassan, M., An, H. K., & **Sameer, M.** (2025). Socioeconomic Mobility and Urban Travel Patterns in Post-COVID-19 United States: Insights from the 2022 National Household Travel Survey. Elsevier. https://doi.org/10.2139/ssrn.5192486
- **Sameer, M.**, Soncco Sinchi, K. W., Amuda, K. A. (2026). Human-Centered Data Fusion and Multi-Scale Digital Twins for Community Resilience Under Multi-Hazard Events. (Preprint)

Technical Reports/ Conferences Attended

- Impact of Modeling Resolution on Flood Risk Analysis Outcomes (2025). EMI 2025. **Sameer, M.**, Nofal, O.
- Ground Investigation & Response of Jhilmil Residential Town Project (2018). ICACE, CUET. Javed, S.A., **Sameer, M.** et al.
- Liquefaction Potentiality Analysis of Jhilmil Residential Town Project (2016). BUET-ANWAR ISPAT Civil Engineering Summit. Javed, S.A., **Sameer, M.**

Framework Development

- NBI FHWA Bridge_Research_Analytics_Scoring
- Florida Statewide Bridge Network — Live Platform
- Selected to Advance to Full Submission Stage, **ASCE Global Innovate Competition: AI & Generative Models Challenge** for Civil and Construction Engineering, 2026.

HONORS, AWARDS & RECOGNITIONS

- FES Florida Professional Engineers in Government (FPEG) Scholarship (**2026**)
- FWEA FE Sponsorship (**2026**)
- Barnes & Noble Graduate Scholarship, FIU (2025, **2026**)
- 1st Prize – Flood Guard Miami, EmpowHer MiniHack (2025)
- Academic Merit Scholarship, King Fahd University of Petroleum and Minerals (KFUPM) (2022–2023)
- Sigma Xi Research Honor Society Member (2025)
- Licensed Engineer, Institute of Engineers Bangladesh (2022)
- Graduate Writing Mentorship Fellow, FIU (2025)

SKILLS

Programming: Python | R | MATLAB
Simulation: CARLA, Unreal Engine4 | SUMO | AIMSUN | VISSIM | Synchro
GIS & Design: ArcGIS | QGIS | AutoCAD | Civil 3D | AutoTURN

LEADERSHIP

Leadership

- RSO Representative, Florida Engineering Society (FES), academic year 2024–**Present**
- Secretary, Bangladesh Student Organization (BSO), FIU, 2025–2026
- Active Graduate Member, ASCE FIU Student Chapter, 2024–2025
- Active Member, ASCE WSU Student Chapter, 2023–2024
- RSO Representative, Florida Water Environment Association (FWEA), 2026–**Present**
- Facilitator, Global Talent Accelerator (GTA), Global Detroit, 2023

Volunteer Experience

- NHERI/Natural Hazard Engineering Research Webinar & Technical Activities, FIU Research, 2025–Present
- Michigander Scholars Exchange Program, 2024